

Programme : Diploma in Cloud Computing and Big data	
Course Code : 5277	Course Title: Machine Learning Lab
Semester : 5	Credits: 1.5
Course Category: Programme core course	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Exposure to Data Preparation or Preprocessing techniques
- Exposure to the implementation of Machine Learning Algorithms for prediction, clustering and classification

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Problem Analysis and Programming concepts	2131	Problem Solving and Programming	2
Programming in Python	3279	Programming in Python Lab	3
Implementation of data structures	4278	Data Structures and Algorithms Lab	4

Course Outcomes:

On completion of the course student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Configure Development Environment, Practice various data preprocessing techniques	12	Applying
CO2	Implement Prediction Algorithm	6	Applying
CO3	Practice Clustering Algorithm	6	Applying

CO4	Construct Classification Algorithms	15	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	3					
CO3	3	3					
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Configure Development Environment, Practice various data preprocessing techniques		
M1.01	Demonstrate working of Google Co-laboratory (Colab) Install Jupyter Notebook or Pytorch development environment	2	Applying
M1.02	Practice Python libraries for Machine Learning: Numpy, Scikit Learn, Pandas, Matplotlib, Tensorflow, Keras.	3	Applying
M1.03	Experiment data cleaning, Missing data handling, Encoding categorical data, Splitting dataset into training and test set, Feature scaling, Data visualization.	4	Applying
M1.04	Apply Principal Component Analysis (PCA) for Dimensionality Reduction.	3	Applying

CO2	Implement Prediction Algorithm		
M2.01	Practice Prediction technique using Linear Regression	6	Applying
	Lab Test – I	3	
CO3	Practice Clustering Algorithm		
M3.01	Apply K-Means clustering	6	Applying
CO4	Construct Classification Algorithms		
M4.01	Practice classification using Logistic Regression	4	Applying
M4.02	Implement classification using Decision Tree and Random Forest	4	Applying
M4.03	Perform comparison of Performance of classification algorithms on the same data using Performance metrics	3	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	Manaranjan Pradhan, U Dinesh Kumar, "Machine Learning using Python", Wiley
T2	Mark Fenner, "Machine Learning with Python for Everyone", Pearson
R3	Pratap Dangeti, "Statistics for Machine Learning", Packt

Online Resources

Sl.No	Website Link
1	https://www.kaggle.com/datasets

2	https://developers.google.com/machine-learning/intro-to-ml
3	vhttps://keras.io/api/datasets/
4	https://www.geeksforgeeks.org/datasets-in-keras